

Assessment Tool C03-AT-2

SIMATS ENGINEERING

Course Name: Computer vision

Course Code: JTA0508

I N. Harshith / 192325078 certify that the submission is my original work and that this submission work is not from any ai tool. I have adhered to the guidelines integrity rules will result in disciplinary action.

Rubrics

Criteria.	weightage	level 4	level 3	level 2	level 1
Understanding of case	25 22	Thorough Understanding	Good Understanding	Basic Understanding	Poor
Application of concepts.	25 20	Effective applies.	Appropriate applies.	Limited applies	Fails to apply
Analysis.	20 15	In-depth analysis.	Good analysis.	Basic Analysis	weak Analysis.
Solution Design	20 15	Innovative feasible	logical solutions.	Basic solutions.	Incorrect solutions
clarity	10 8	clear, well structured.	organized presentation	limited clarity	Poor clarity.

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Case Study

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO3: *Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*

Assessment Tool Weightage: 20%

Read the given scenario carefully and analyze the problem using appropriate computer vision techniques. Justify your answers with relevant reasoning and comparisons wherever necessary.

1. Preprocessing and Feature Detection

A vision system processes images captured under varying lighting conditions. Without preprocessing, feature detection results are inconsistent.

Analyze the issue and explain how preprocessing improves feature detection performance.

1. Preprocessing and Feature Detection

Case: A vision system processes images captured under varying lighting conditions. Without preprocessing, feature detection results are inconsistent.

Analytical Answer

The main problem is that **varying lighting conditions change the intensity and contrast of an image**. As a result, the same object may appear different in different images. Feature detectors may then detect different keypoints or fail to detect important features, making the system inconsistent.

Preprocessing helps make the images more uniform before feature detection. Techniques such as **normalization, contrast enhancement, and noise removal** can reduce the effect of illumination changes.

For example:

- **Normalization** adjusts the intensity range.
- **Contrast enhancement** makes important image structures more visible.
- **Noise removal** eliminates unwanted variations that may produce false features.

- **Histogram equalization** can improve the visibility of features in poorly illuminated images.

After preprocessing, feature detection methods such as **SIFT or Harris Corner Detection** can identify more stable and meaningful features.

Conclusion

Therefore, preprocessing acts as an important initial stage in the vision pipeline. It reduces variations caused by lighting and noise, resulting in **more consistent, reliable, and accurate feature detection**. This improves the performance of later tasks such as **feature matching and image comparison**.

2. Edge vs Corner Detection

In an image containing both straight edges and textured regions, a system uses edge detection but fails to identify key points for matching.

Analyze why edge detection is insufficient and justify the use of corner detection in this case.

2. Edge vs Corner Detection

Case: An image contains both straight edges and textured regions. The system uses edge detection but fails to identify key points for matching.

Analytical Answer

The problem is that **edge detection identifies boundaries**, but it does not necessarily provide **distinctive key points** that can be reliably used for feature matching.

An edge represents a location where the image intensity changes significantly. For example, the boundary of an object can be detected as an edge. However, an edge can extend over a long distance, so it does not provide a unique point for matching between two images.

Corner detection is more suitable in this situation because corners are points where intensity changes significantly in **multiple directions**. They provide distinctive locations that can be used as key points.

For the given image:

- **Edge detection** → identifies straight boundaries and object contours.
- **Corner detection** → identifies distinctive points in textured regions.
- These corner points can then be used for **feature description and matching** between images.

Therefore, using only edge detection causes the system to miss important key points required for reliable matching.

Conclusion

Edge detection is useful for identifying **object boundaries**, but it is insufficient when the objective is to find distinctive points for matching. **Corner detection should be used along with or instead of edge detection** because corners provide more localized and distinctive features, improving the reliability of feature matching.

3. Hough Transform in Noisy Images

A system uses Hough Transform to detect lines in noisy images but produces inaccurate results. Analyze the effect of noise on detection and suggest improvements based on preprocessing or parameter selection.

3. Hough Transform in Noisy Images

Case: A system uses Hough Transform to detect lines in noisy images but produces inaccurate results.

Analytical Answer

The main problem is that **noise creates unwanted edges and intensity variations** in the image. When the Hough Transform is applied to such an image, these unwanted edges may also contribute votes in the Hough space. As a result, the system may detect **false or inaccurate lines**.

To improve the detection, preprocessing should be performed before applying the Hough Transform. **Noise removal using Gaussian or median filtering** can reduce unwanted variations. After filtering, an edge detector such as **Canny** can be applied to obtain cleaner edges.

The Hough Transform parameters should also be selected properly. Important parameters include:

- **Threshold** – controls how many votes are required to detect a line.
- **Minimum line length** – removes very short unwanted lines.
- **Maximum line gap** – determines how much gap can exist between connected line segments.

Using appropriate parameter values helps distinguish meaningful lines from noise.

Conclusion

Noise negatively affects Hough Transform by producing false edge points and incorrect line detections. A better approach is:

Noise Removal → Edge Detection → Parameter Tuning → Hough Transform

Thus, proper preprocessing and suitable Hough parameters improve **line detection accuracy and reliability** in noisy images.

4. Feature Matching under Transformations

Two images of the same object are taken at different scales and orientations. Traditional feature matching fails, but scale invariant feature matching works effectively. Analyze why scale invariant methods perform better in this scenario.

4. Feature Matching under Transformations

Case: Two images of the same object are taken at different scales and orientations. Traditional feature matching fails, but scale-invariant feature matching works effectively.

Analytical Answer

The main reason traditional feature matching fails is that the appearance and position of image features change when the object is **scaled or rotated**. A feature that appears at one size or orientation may not have the same representation in the second image.

Scale-invariant feature matching is designed to handle such transformations. It detects distinctive features and creates descriptors that remain relatively stable when the image is changed in scale or orientation.

For example, **SIFT** identifies keypoints at different scales and generates descriptors based on the local image structure. Therefore, even when the same object appears larger, smaller, or rotated, corresponding features can still be identified and matched.

The process can be represented as:

Image → Keypoint Detection → Feature Description → Feature Matching

Scale-invariant methods are more effective because the descriptors are designed to preserve important feature information despite changes in **scale and orientation**.

Conclusion

Traditional feature matching is sensitive to transformations such as scaling and rotation. **Scale-invariant methods such as SIFT** provide more robust feature representations, allowing corresponding features to be matched even when the same object appears at different sizes and orientations. This makes them more suitable for reliable image comparison and object recognition.

5. PCA for Feature Optimization

A system extracts a large number of features, leading to high computational cost and redundancy.

Analyze how Principal Component Analysis helps in reducing dimensionality and improving efficiency.

5. PCA for Feature Optimization

Case: A system extracts a large number of features, leading to high computational cost and redundancy.

Analytical Answer

The main problem is that extracting a large number of features creates a **high-dimensional feature space**. Many of these features may contain redundant or less useful information. Processing all the features increases **computational cost, memory usage, and processing time**.

Principal Component Analysis (PCA) helps solve this problem by reducing the number of dimensions while preserving the most important information in the original feature set.

PCA works by:

1. Taking the original high-dimensional features.
2. Finding the directions with the greatest variance.
3. Ranking these directions according to their importance.
4. Selecting a smaller number of principal components.
5. Representing the original features using these components.

For example, if the original feature vector contains **1000 features**, PCA may represent it using **100 important components**. This reduces the amount of data that needs to be processed.

The main benefits are:

- Reduces feature dimensionality.
- Removes redundant information.
- Reduces computational cost.
- Reduces memory requirements.
- Makes subsequent processing faster.

However, selecting too few components can remove useful information. Therefore, the number of components should be selected carefully to maintain sufficient information.

Conclusion

PCA provides an effective way to optimize a large feature set by converting it into a smaller set of important components. This reduces **computational complexity and redundancy** while retaining the most significant information, making feature-based computer vision systems more efficient.

6. Preprocessing Impact on Edge Detection

A system performs edge detection on raw images containing noise and uneven illumination, resulting in fragmented edges. Analyze the problem and explain how preprocessing improves edge detection accuracy and continuity.

6. Preprocessing Impact on Edge Detection

Case: A system performs edge detection on raw images containing noise and uneven illumination, resulting in fragmented edges.

Analytical Answer

The main problem is that **noise and uneven illumination affect the intensity values of the image**. Edge detection algorithms depend on intensity changes, so noise can create false edges while uneven lighting can make real edges weak or discontinuous. This results in **fragmented and inaccurate edges**.

Preprocessing should therefore be applied before edge detection. **Noise removal** using smoothing filters such as Gaussian filtering can reduce unwanted intensity variations. **Normalization or contrast enhancement** can reduce the effect of uneven illumination and make important boundaries more distinguishable.

The improved process can be represented as:

Raw Image → Noise Removal → Intensity/Contrast Normalization → Edge Detection

After preprocessing, the edge detector receives a cleaner and more consistent image. This helps connect broken edge regions and reduces false edge responses.

Conclusion

Preprocessing improves edge detection by **reducing noise and correcting intensity variations** caused by uneven illumination. As a result, the detected edges become **clearer, more continuous, and more accurate**, improving the reliability of subsequent computer vision tasks.

7. Corner Detection vs Edge Detection in Tracking

A tracking system uses edge detection but fails to track moving objects accurately across frames.

Analyze why edges are insufficient and justify the use of corner detection for reliable tracking.

7. Corner Detection vs Edge Detection in Tracking

Case: A tracking system uses edge detection but fails to track moving objects accurately across frames.

Analytical Answer

The main problem is that **edge detection identifies boundaries or contours**, but edges are often long and do not provide unique points that can be reliably tracked from one frame to another.

For object tracking, the system needs **distinctive and stable points** that can be identified across consecutive frames. **Corner detection** is more suitable because corners represent locations where intensity changes significantly in multiple directions.

For example:

- **Edge detection** → identifies object boundaries.
- **Corner detection** → identifies distinctive key points.
- These key points can be matched across consecutive frames to estimate object movement.

Using corner points also makes it easier to determine how an object has moved because the same corner can be located in successive frames.

Conclusion

Edge detection alone is insufficient for accurate object tracking because it mainly provides boundaries rather than distinctive tracking points. **Corner detection provides localized and distinctive key points**, making it more suitable for reliable tracking of moving objects across frames.

8. Hough Transform for Shape Detection

A system uses Hough Transform to detect circular shapes but fails in complex backgrounds. Analyze the limitations and explain how preprocessing or parameter tuning can improve detection.

8. Hough Transform for Shape Detection

Case: A system uses Hough Transform to detect circular shapes but fails in complex backgrounds.

Analytical Answer

The main problem is that a **complex background contains many unwanted edges and patterns**. When the Hough Transform is applied directly, these edges can generate incorrect votes in the Hough space. As a result, the system may detect **false circles** or fail to identify the actual circular shape.

To improve the detection, preprocessing should be performed before applying the Hough Transform. **Noise removal and smoothing** can reduce unwanted edges. Edge detection can then be applied to obtain cleaner boundaries.

The Hough Circle Transform parameters should also be properly tuned. Important parameters include:

- **Minimum distance between circles** – prevents multiple detections of the same circle.
- **Edge detection threshold** – controls the sensitivity of circle detection.
- **Accumulator threshold** – determines how strong the evidence must be for a circle to be detected.
- **Minimum and maximum radius** – restricts detection to circles of the expected size.

The improved process can be represented as:

Input Image → Noise Removal → Smoothing → Edge Detection → Parameter Tuning → Hough Circle Transform

Conclusion

The Hough Transform can produce inaccurate circular detections in complex backgrounds because unwanted edges interfere with the voting process. **Preprocessing and proper parameter selection** reduce false detections and improve the accuracy and reliability of circular shape detection.

9. Feature Matching under Illumination Changes

Two images of the same scene are captured under different lighting conditions, causing feature matching errors. Analyze why matching fails and explain how preprocessing or robust descriptors improve performance.

9. Feature Matching under Illumination Changes

Case: Two images of the same scene are captured under different lighting conditions, causing feature matching errors.

Analytical Answer

The main problem is that **changes in illumination alter the intensity, brightness, and contrast of image regions**. Although the scene is the same, the appearance of its features may be different in the two images. As a result, feature descriptors may not be sufficiently similar, causing incorrect or missed matches.

To improve matching, **preprocessing** can be applied before feature extraction. Techniques such as **normalization and contrast enhancement** can reduce the effect of lighting differences and produce a more consistent image representation.

Another solution is to use **robust feature descriptors** that are less sensitive to illumination changes. Such descriptors focus more on local structural information rather than only raw pixel intensity.

The improved process can be represented as:

Input Images → Preprocessing → Feature Detection → Robust Feature Description → Feature Matching

Conclusion

Illumination changes can make corresponding features appear different and therefore reduce matching accuracy. Applying suitable preprocessing and using **robust feature descriptors** can reduce the influence of lighting variations and improve the **reliability and accuracy of feature matching**.

10. PCA in Face Recognition

A face recognition system processes high-dimensional image data, resulting in slow performance.

Analyze how PCA helps in reducing dimensionality while preserving important features.

10. PCA in Face Recognition

Case: A face recognition system processes high-dimensional image data, resulting in slow performance.

Analytical Answer

The main problem is that face images contain a **large number of pixels**, resulting in highdimensional data. Processing all these dimensions increases **computation time and memory requirements**, making the face recognition system slower.

Principal Component Analysis (PCA) can be used to reduce the dimensionality of the face image data. PCA transforms the original high-dimensional features into a smaller number of **principal components** that preserve the most important variations in the data.

For example, instead of processing thousands of original pixel features, the system can represent a face using a much smaller number of principal components. This reduces the amount of data that the recognition algorithm needs to process.

The process can be represented as:

Face Image → Feature Representation → PCA → Reduced Features → Face Recognition

The main advantages are:

- Reduces the number of dimensions.
- Decreases computational cost.
- Reduces memory usage.
- Speeds up face recognition.
- Removes some redundant information.

However, if too few components are selected, important facial information may be lost. Therefore, the number of components should be selected carefully.

Conclusion

PCA improves face recognition efficiency by reducing high-dimensional image data into a smaller feature representation while preserving important information. This decreases computational complexity and makes the recognition system faster.

11. Edge Detection in Low Contrast Images

An image has low contrast, causing edge detection algorithms to fail. Analyze the problem and explain how preprocessing enhances edge detection results.

11. Edge Detection in Low Contrast Images

Case: An image has low contrast, causing edge detection algorithms to fail.

Analytical Answer

The main problem is that a **low-contrast image has only small differences in intensity between neighboring pixels**. Since edge detection algorithms depend on intensity changes, weak boundaries may not produce sufficiently strong responses. As a result, important edges may be missed.

To improve the result, **preprocessing can be applied before edge detection**. Contrast enhancement increases the difference between dark and bright regions, making object boundaries more visible. **Normalization** can also improve the intensity distribution of the image.

The improved process can be represented as:

Low-Contrast Image → Contrast Enhancement/Normalization → Edge Detection

After enhancement, the intensity difference around object boundaries becomes stronger, allowing the edge detector to identify more meaningful edges.

Conclusion

Low contrast makes edges weak and difficult to detect. Applying **contrast enhancement and normalization before edge detection** improves edge visibility and helps produce **clearer and more complete edge information**.

12. Corner Detection in Textured Regions

A system applies corner detection in highly textured regions, resulting in excessive keypoints. Analyze the issue and suggest methods to improve feature selection.

12. Corner Detection in Textured Regions

Case: A system applies corner detection in highly textured regions, resulting in excessive keypoints.

Analytical Answer

The main problem is that **highly textured regions contain many intensity changes and small structures**. When corner detection is applied, the algorithm identifies a large number

of these variations as possible corners. This produces **excessive keypoints**, including some that may not be useful for matching or tracking.

Too many keypoints can increase **computational cost** and may also make feature matching less efficient because the system has to compare a larger number of features.

To improve feature selection, the system can use:

- **Thresholding** to retain only strong corner responses.
- **Non-maximum suppression** to remove nearby or weaker keypoints.
- **Feature ranking** to select the most distinctive points.
- **Limiting the number of keypoints** to the strongest detected features.

The improved process can be represented as:

Textured Image → Corner Detection → Keypoint Filtering → Strong Keypoints → Feature Matching

Conclusion

Highly textured regions naturally produce many corner responses, but not all of them are useful. Therefore, **keypoint filtering, thresholding, and selection of the strongest features** can reduce excessive keypoints while retaining important information. This improves computational efficiency and makes subsequent feature matching more reliable.

13. Feature Matching with Noise

Feature matching between two images fails due to noise in one image. Analyze the impact of noise and explain how preprocessing improves matching accuracy.

13. Feature Matching with Noise

Case: Feature matching between two images fails due to noise in one image.

Analytical Answer

The main problem is that **noise introduces unwanted variations in the image**. These variations can create false features or change the appearance of existing features. As a result, the feature descriptors generated from the noisy image may not match correctly with the corresponding features in the other image.

To improve matching accuracy, **preprocessing should be performed before feature extraction**. Noise removal techniques such as **Gaussian filtering or median filtering** can reduce unwanted variations while preserving important image structures.

The improved process can be represented as:

Noisy Image → Noise Removal → Feature Detection → Feature Description → Feature Matching

After preprocessing, the extracted features become more stable and meaningful, increasing the possibility of finding correct correspondences.

Conclusion

Noise can reduce feature matching accuracy by introducing false features and changing existing feature descriptions. Applying **appropriate noise removal during preprocessing** helps preserve important image structures and improves the **accuracy and reliability of feature matching**.

14. PCA and Information Loss

A system applies PCA aggressively and loses important feature details. Analyze the trade-off between dimensionality reduction and information loss, and justify optimal component selection.

14. PCA and Information Loss

Case: A system applies PCA aggressively and loses important feature details.

Analytical Answer

The main problem is that **too many dimensions are removed during PCA**. Although PCA reduces the size of the feature data, selecting very few principal components can discard important information. This causes **information loss** and may reduce the accuracy of later computer vision tasks.

There is therefore a trade-off between:

- **Dimensionality reduction** → lower computational cost and faster processing.
- **Information preservation** → retaining important feature characteristics.

If too few components are selected, important variations in the original features may be lost. If too many components are retained, the computational advantage of PCA becomes smaller.

The number of components should therefore be selected based on the amount of **explained variance** that needs to be preserved. The system can gradually increase the number of components and choose a value that provides sufficient information while still achieving meaningful dimensionality reduction.

The process can be represented as:

Original Features → PCA → Select Appropriate Components → Reduced Features

Conclusion

PCA should not be applied too aggressively because excessive dimensionality reduction can remove important feature information. An **optimal number of principal components** should be selected by balancing computational efficiency with information preservation. This provides a reduced representation while maintaining the important characteristics required for accurate computer vision processing.

15. Combined Feature Detection and Matching

A system detects features correctly but fails during matching due to incorrect descriptor selection.

Analyze the issue and explain how appropriate feature descriptors improve matching performance.

15. Combined Feature Detection and Matching

Case: A system detects features correctly but fails during matching due to incorrect descriptor selection.

Analytical Answer

The main problem is that **feature detection and feature description are different stages**. A detector may successfully identify important keypoints, but if an unsuitable descriptor is used, the keypoints cannot be represented effectively for comparison. This results in poor or incorrect feature matching.

A **feature descriptor** represents the local characteristics around a detected keypoint. The descriptor should be selected according to the requirements of the application and the type of image transformation expected.

For example, if the images may contain **scale and orientation changes**, a robust descriptor such as the one associated with **SIFT** can provide a more suitable representation for matching.

The improved process can be represented as:

Image → Feature Detection → Appropriate Descriptor → Descriptor Matching → Verified Matches

The descriptor should provide distinctive information so that corresponding features in two images can be correctly identified.

Conclusion

Correct feature detection alone does not guarantee successful matching. The system must use an **appropriate feature descriptor** that represents the detected keypoints effectively. Selecting a suitable descriptor improves **matching accuracy, reliability, and robustness** under image variations.

Code of Conduct:

*I N.Harshith/192325078 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts

Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand